EXPLICIT INVOLUTION GEOMETRY OF PRODUCT KUMMER SURFACES

DEEP BHATTACHARJEE AND PALLAB NANDI

Abstract. We classify the 31 nonidentity involutions in the standard affine two-torsion subgroup of a product Kummer surface. We compute their fixed loci, quotient surfaces and rational cohomology representations, including the fixed loci of all fifteen translations descending to each of the nine Lieberman quotients.

1. Introduction and scope

Let E and F be complex elliptic curves, let $A = E \times F$, and let $X = \mathrm{Km}(A)$ be the minimal resolution of $A/\langle -1\rangle$. The sixteen translations by points of $A[2]$ act on X. The reflection $r(x, y) = (x, -y)$ acts as well. Together they form a particularly accessible subgroup of $\mathrm{Aut}(X)$: every element is given by a formula on A, while the exceptional curves can be tracked through the sixteen points of $A[2]$.

This article gives a complete calculation inside that subgroup. The classification holds without a genericity assumption on E and F. Extra automorphisms may occur for special elliptic curves, but they do not alter the fixed loci of the specified maps. We also calculate the action on rational cohomology and the induced translations on the Enriques quotients.[1] The proofs use the explicit local resolution, elementary equations on elliptic curves, and finite-dimensional linear algebra.

The free involutions in this construction are the classical Lieberman involutions. They are discussed explicitly by Ohashi [4]. Our classification concerns the indicated affine subgroup, not all involutions of a Kummer surface. In particular, the nine free elements below are not a classification of all Enriques quotients. A generic product Kummer surface has fifteen distinct Enriques quotients up to isomorphism [4]. Counting elements of a group and counting isomorphism classes of quotient surfaces are different tasks.

The paper replaces several inaccurate assertions in the earlier review [5]. A diagonal involution of the Fermat quartic that was described as free has eight fixed points. A symplectic involution on a general K3 surface has a K3 resolution as quotient, but the resolution need not be Kummer. Positive Picard rank alone is not the projectivity criterion. These distinctions are proved or explained in Sections 7–8 and Appendix B. The revised title describes the explicit family treated here.

1.1. **Conventions.** All varieties and morphisms are over $\mathbb{C}$. A K3 surface means a smooth projective surface with trivial canonical bundle and $H^1(\mathcal{O}) = 0$. An Enriques surface means a smooth projective surface with $h^{1,0} = h^{2,0} = 0$ and canonical bundle of exact order two. Intersection forms are integral unless coefficients are stated. The lattice U has matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, and $E_8(-1)$ is negative definite. We write $L(m)$ for a lattice with its form multiplied by m. The elementary surface theory used below is covered by Beauville [1] and Huybrechts [2].

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Deep Bhattacharjee is the corresponding author. ORCID: 0000-0003-0466-750X..

[1]The nine free group elements need not give nine pairwise nonisomorphic quotient surfaces for special choices of the elliptic curves. No such isomorphism classification is used here.

1.2. **The two principal statements.** Put $T = A[2]$, write t_c for translation by c, and set $G = \langle T, r\rangle$. In the following table, $a \in E[2]$ and $b \in F[2]$.

Theorem 1.1. *The induced action identifies G with $(\mathbb{Z}/2\mathbb{Z})^5$. Its 31 nonidentity elements have the following fixed loci on X.*

Element	*Condition*	*Number*	*Fixed locus*
t_c	$c \neq 0$	*15*	*eight points*
$rt_{(a,b)}$	$a \neq 0$, $b \neq 0$	*9*	*empty*
$rt_{(a,0)}$	$a \neq 0$	*3*	*two copies of F*
$rt_{(0,b)}$	$b \neq 0$	*3*	*two copies of E*
r		*1*	*eight rational curves*

The translations are symplectic and the other sixteen elements are antisymplectic. Resolving $X/\langle t_c\rangle$, for $c \neq 0$, gives $\mathrm{Km}(A/\langle c\rangle)$. *Each of the nine free quotients is Enriques.*

Theorem 1.2. *Fix $a \neq 0$ and $b \neq 0$, set $\sigma = rt_{(a,b)}$, and let $S = X/\langle\sigma\rangle$. All sixteen translations descend faithfully to S. Every nonidentity descended translation has four isolated fixed points. Among these fifteen maps, one has four additional rational curves, two have two additional elliptic curves, four have one additional elliptic curve, and eight have no fixed curve. The element associated with (a, b) acts trivially on $H^2(S, \mathbb{Q})$; each of the other fourteen has trace two.*

The fixed-point calculations prove these statements for every product $E \times F$. The accompanying exact-arithmetic program checks the finite group bookkeeping and character multiplicities. It is a check of the finite combinatorics, not a replacement for the geometric arguments.

2. Constructing the Kummer surface

2.1. **The sixteen singular points.** Write $\kappa(p) = -p$. The fixed locus of κ on A is $A[2]$, which has sixteen points. At any such point choose analytic coordinates (z, w) in which $\kappa(z, w) = (-z, -w)$. The invariant local coordinates z^2, zw, w^2 satisfy one relation,

$$(zw)^2 = z^2w^2.$$

Thus the quotient has an ordinary double point of type A_1. Away from these points the quotient map is an unramified double covering.

Let $b : \widetilde{A} \to A$ be the blowup of the sixteen points. In one blowup chart write $z = u, w = uv$. The lifted involution acts by

$$(u, v) \longmapsto (-u, v).$$

The quotient has coordinates $(t, v) = (u^2, v)$ and is smooth. The other chart gives the same conclusion. It follows that

$$X = \widetilde{A}/\langle\kappa\rangle$$

is the minimal resolution of $A/\langle\kappa\rangle$. Each exceptional curve on X is a rational curve of self-intersection -2. Indeed, the exceptional curve on $\widetilde{A}$ has square -1; its pullback as a divisor from X is twice that curve. The degree-two intersection formula therefore gives $2C^2 = 4(-1)$.

2.2. **The two-form and the invariants.** Choose nonzero holomorphic one-forms on E and F. Their exterior product ω is nowhere zero on A and invariant under κ. In the coordinates just used,

$$dz \wedge dw = u\,du \wedge dv = \frac{1}{2}\,dt \wedge dv.$$

Consequently the descended two-form extends across the exceptional curve without a zero. This proves $K_X \simeq \mathcal{O}_X$ directly.

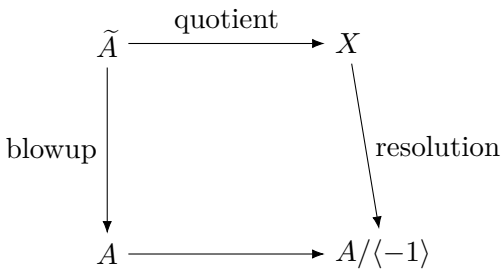


FIGURE 1. The smooth Kummer model. The upper horizontal map is branched along sixteen exceptional curves. The lower quotient has sixteen nodes.

Blowing up points does not change $H^1(\mathcal{O})$. The involution κ acts as -1 on $H^1(A, \mathcal{O}_A)$, so its invariant part vanishes. Taking invariants under a finite group in characteristic zero is exact; applying this to the finite quotient shows $H^1(X, \mathcal{O}_X) = 0$. Projectivity follows from the projective blowup and finite quotient. Hence X is K3.

The Euler characteristic of A is zero. Each point blowup adds one, so $e(\widetilde{A}) = 16$. The fixed locus of the lifted κ is the disjoint union of sixteen projective lines and has Euler characteristic 32. The elementary Euler formula for an involution gives

$$e(X) = \frac{16 + 32}{2} = 24.$$

Since $b_1 = b_3 = 0$, it follows that $b_2 = 22$.

Remark 2.1. The local calculation is also why the singular quotient and its smooth resolution must not be interchanged. A line bundle pulled back from the singular quotient has intersection zero with a contracted exceptional curve, so that pullback cannot be ample on the resolution.

3. The affine two-torsion subgroup

3.1. **Descent and faithfulness.** For $c \in T$, the equality $2c = 0$ implies $t_c\kappa = \kappa t_c$. Also $r\kappa = \kappa r$. These maps permute the centers of the blowup and therefore lift to $\widetilde{A}$ and descend to X. On T, the reflection r is the identity. Hence r commutes with all t_c, and all resulting elements have order at most two.

No nonzero translation induces the identity on X. If it did, then on the complement of the nodes one would have $p + c = p$ or $p + c = -p$ for every p. The first equality forces $c = 0$; the second describes only the finite set $2p = -c$. Similarly, a map rt_c cannot induce the identity: neither $rt_c = p$ nor $rt_c = -p$ can hold on an open subset of the two-dimensional abelian surface. Thus the 32 maps are distinct.

Translations preserve ω, whereas $r^*\omega = -\omega$. Their symplectic types are unchanged on passing to X.

3.2. **Fixed points of a translation.**

Proposition 3.1. *For $c \neq 0$, $\mathrm{Fix}_X(t_c)$ consists of eight points outside the exceptional curves. The derivative at each of these points is $-I$.*

Proof. A nonexceptional point of X is the orbit $\{p, -p\}$. It is fixed precisely when $p + c = p$ or $p + c = -p$. The first alternative is impossible. The second is $2p = -c = c$. Multiplication by two on a complex abelian surface has degree sixteen, so it has sixteen solutions. None belongs to $A[2]$, since $c \neq 0$. Pairing p with $-p$ leaves eight orbits. Translation acts on $A[2]$ without a fixed point, so it fixes no exceptional curve or point on such a curve.

Near a solution, identify the neighborhood of $-p$ with that of p using inversion. The composite with translation then has derivative $-I$. In particular all eight points are isolated and nondegenerate. □

Proposition 3.2. *Let $B = A/\langle c\rangle$, $c \neq 0$. The minimal resolution of $X/\langle t_c\rangle$ is* $\mathrm{Km}(B)$.

Proof. The commuting actions yield an isomorphism of normal quotient surfaces

$$A/\langle \kappa, t_c\rangle \simeq (A/\langle c\rangle)/\langle -1\rangle.$$

The surface B is abelian, since the translation quotient is a finite isogeny. Its Kummer resolution is smooth with trivial canonical bundle by the calculation of Section 2. The quotient $X/\langle t_c\rangle$ has only the eight A_1 singularities of Proposition 3.1; its resolution also has trivial canonical bundle and $h^{1,0} = 0$. Both smooth resolutions are birational to the displayed normal quotient. Smooth minimal K3 surfaces that are birational are isomorphic: a birational map between smooth surfaces factors into point blowups and blowdowns, whereas a K3 surface has no (-1)-curve by adjunction. This proves the assertion. □

The product hypothesis enters the reflection calculation below, but not Proposition 3.2. In particular B need not itself be a product of elliptic curves.

4. The sixteen antisymplectic involutions

Set $\sigma_{a,b} = rt_{(a,b)}$. On A this is

$$(x, y) \longmapsto (x + a, -y + b).$$

For a point outside $A[2]$, its orbit in the quotient is fixed exactly when one of the following equations holds:

$$\sigma_{a,b}(x, y) = (x, y) \quad \Longleftrightarrow \quad a = 0, \quad 2y = b, \tag{1}$$

$$\sigma_{a,b}(x, y) = (-x, -y) \quad \Longleftrightarrow \quad b = 0, \quad 2x = a. \tag{2}$$

These two equations exhaust the possibilities, since the quotient orbit has only two elements.

4.1. **Both translation components nonzero.** If a and b are nonzero, neither equation can hold. On the sixteen nodes, $\sigma_{a,b}$ acts as translation by (a, b), so it pairs the nodes and their exceptional curves without a fixed point. Therefore $\sigma_{a,b}$ is free on all of X. There are $3 \cdot 3 = 9$ choices.

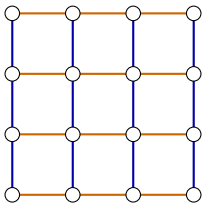
4.2. **Exactly one translation component nonzero.** Suppose $a \neq 0$ and $b = 0$. Equation (2) gives four roots of $2x = a$, and y is arbitrary. None of these x lies in $E[2]$. Inversion pairs the four roots into two pairs. On each pair of copies of F, the quotient by inversion is isomorphic to one copy of F: choosing one root identifies that component with F, and the other root is its inverted partner. Thus the fixed locus consists of two disjoint smooth elliptic curves. No exceptional curve is fixed, since the nodes are translated by the nonzero point $(a, 0)$. The case $a = 0, b \neq 0$ follows from Equation (1) and gives two copies of E.

4.3. **The reflection itself.** For $a = b = 0$, Equations (1)–(2) give

$$E \times F[2] \quad \text{and} \quad E[2] \times F.$$

After quotienting by inversion, each of these eight elliptic curves becomes a projective line. Their proper transforms on X are disjoint. At a crossing, the two branches meet the exceptional curve at distinct points, separating the proper transforms.

To check that no further fixed point has been missed, use local coordinates with $r(z, w) = (z, -w)$. In the blowup chart $z = u, w = uv$, the action is $(u, v) \mapsto (u, -v)$. On the quotient chart $(t, v) = (u^2, v)$ it remains $(t, v) \mapsto (t, -v)$. The exceptional curve $t = 0$ has one fixed point in this



four vertical and four horizontal curves
sixteen resolved crossings

FIGURE 2. Incidence schematic of the double Kummer configuration. Each small circle represents an exceptional curve, not a remaining intersection of the eight fixed rational curves.

chart, $v = 0$, already belonging to the proper transform of $w = 0$. The other chart gives the other fixed point, on the proper transform of $z = 0$. There are no isolated fixed points left over.

4.4. **The Enriques quotients.** Let $\sigma = \sigma_{a,b}$ with $a, b \neq 0$. The map $\pi : X \to S = X/\sigma$ is an unramified double covering of smooth projective surfaces. Since σ acts by -1 on $H^0(X, K_X)$, we have $h^{2,0}(S) = 0$. Also $H^1(S, \mathcal{O}_S)$ injects into $H^1(X, \mathcal{O}_X) = 0$. The covering formula gives $\pi^* K_S = K_X = 0$; taking the norm gives $K_S^{\otimes 2} \simeq \mathcal{O}_S$. It is not trivial, since otherwise $h^{2,0}(S) = 1$. Hence S is Enriques.

The topological covering gives $e(S) = 12$, and Hodge theory gives $b_1 = b_3 = 0$, so $b_2(S) = 10$. For divisor classes D, D',

$$(\pi^* D, \pi^* D')_X = 2(D, D')_S. \tag{3}$$

The same formula holds for rational degree-two cohomology classes. [2] This finishes the geometric proof of Theorem 1.1.

5. The rational cohomology representation

5.1. **Cohomology through the resolution.** There is a G-equivariant rational decomposition

$$H^2(X, \mathbb{Q}) \simeq H^2(A, \mathbb{Q}) \oplus \bigoplus_{v \in T} \mathbb{Q} e_v. \tag{4}$$

One way to see it is to apply the blowup formula on $\widetilde{A}$. The lifted κ acts trivially on $H^2(A, \mathbb{Q})$, because it acts as -1 on $H^1(A, \mathbb{Q})$, and it fixes every exceptional class. Rational cohomology of a finite quotient identifies with the invariant cohomology of the covering space. The latter statement follows, for example, by averaging cochains after a compatible triangulation; the averaging operator is available because the group order is invertible in $\mathbb{Q}$. This yields (4).

Translations act trivially on $H^2(A, \mathbb{Q})$ and regularly permute the sixteen e_v. The Künneth decomposition is

$$H^2(A, \mathbb{Q}) = H^2(E, \mathbb{Q}) \oplus \big(H^1(E, \mathbb{Q}) \otimes H^1(F, \mathbb{Q})\big) \oplus H^2(F, \mathbb{Q}).$$

The reflection has signs $+, -, +$ on these three terms, of dimensions $1, 4, 1$. It fixes the exceptional classes.

[2] The factor two is the degree of the unramified covering. It must be retained when comparing the Enriques numerical lattice with its pullback in the K3 lattice.

Proposition 5.1. *Let $\widehat{T} = \mathrm{Hom}(T, \{\pm 1\})$. Write $\mathbb{Q}_{\chi,\epsilon}$ for the character on which t_c acts by $\chi(c)$ and r acts by ϵ. Then*

$$H^2(X, \mathbb{Q}) \simeq \mathbb{Q}_{1,+}^3 \oplus \mathbb{Q}_{1,-}^4 \oplus \bigoplus_{\substack{\chi \in \widehat{T} \\ \chi \neq 1}} \mathbb{Q}_{\chi,+}.$$

Proof. The regular representation of the elementary abelian group T is the sum of its sixteen rational characters, each once. The reflection acts as $+1$ on that entire summand. The Künneth calculation contributes two trivial characters with sign $+$ and four with sign $-$. Adding these contributions proves the formula; the dimensions sum to 22. □

Corollary 5.2. *The traces on $H^2(X, \mathbb{Q})$ are 6 for t_c, $c \neq 0$, 14 for r, and -2 for rt_c, $c \neq 0$. The corresponding invariant dimensions are $14, 18, 10$.*

Proof. A nonidentity translation has trace zero on the regular summand and trace six on $H^2(A)$. The reflection has trace $16 + 2 - 4 = 14$. Combining a nonzero translation with the reflection gives $0 + 2 - 4 = -2$. For an involution on a 22-dimensional space, the invariant dimension is $(22 + \mathrm{tr})/2$. □

The topological Lefschetz formula reads $e(\mathrm{Fix}(g)) = 2 + \mathrm{tr}(g|H^2)$. It gives respectively $8, 16, 0$, agreeing with all of the fixed loci above. In particular the elliptic fixed curves and the empty fixed locus have the same Euler characteristic. A trace calculation alone does not distinguish them; the geometric equations are necessary.

6. Translations on a Lieberman quotient

Fix nonzero $a \in E[2]$, $b \in F[2]$, and put $c_0 = (a, b)$, $\sigma = rt_{c_0}$, $S = X/\sigma$. Every translation commutes with σ and induces an automorphism $\bar{t}_c$ of S. The descended action is faithful: the kernel on X would have to be contained in the deck group $\{1, \sigma\}$, but no translation equals the antisymplectic map σ.

6.1. Lifting the fixed locus.

For any $c \neq 0$,

$$\pi^{-1} \mathrm{Fix}_S(\bar{t}_c) = \mathrm{Fix}_X(t_c) \sqcup \mathrm{Fix}_X(\sigma t_c). \tag{5}$$

The union is disjoint: a point fixed by both would be fixed by σ, contrary to freeness. Each set is σ-stable. The first contains eight points, paired freely by σ, and therefore gives four isolated points on S. They are nondegenerate because the local derivative is $-I$. Any fixed curve comes from the second set. Now

$$\sigma t_c = r t_{c_0 + c}.$$

Thus the previous classification reduces the calculation to the pairing of the relevant elliptic or rational curves.

Theorem 6.1. *Besides four isolated points, the fixed locus of $\bar{t}_c$ has the following curves. Every entry is a disjoint union.*

c	*Number of choices*	*Fixed curves on S*
(a, b)	*1*	*four copies of $\mathbb{P}^1$*
$(a, 0)$	*1*	*two copies of $E/\langle a \rangle$*
$(0, b)$	*1*	*two copies of $F/\langle b \rangle$*
(a, d), $d \notin \{0, b\}$	*2*	*one copy of E*
(d, b), $d \notin \{0, a\}$	*2*	*one copy of F*
all other nonzero c	*8*	*none*

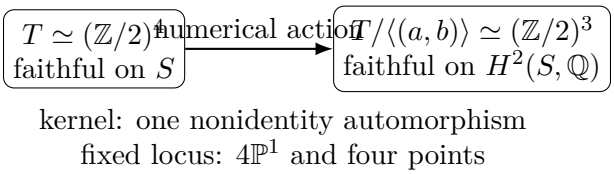


FIGURE 3. Faithfulness on the surface differs from faithfulness on its numerical divisor space. Both kernels are computed explicitly.

Proof. For $c = c_0$, the second set in (5) is the eight rational curves fixed by r. The map σ translates their two-torsion indices by b or a, so it pairs them into four pairs. Each pair descends to one rational curve.

For $c = (a, 0)$, the second set is the fixed locus of $rt_{(0,b)}$, namely two copies of E. They arise from the four roots of $2y = b$, paired by inversion. On these roots, $-y + b = y$, because $2y = b$. Hence σ preserves each resulting elliptic curve and acts on it by $x \mapsto x + a$. The quotient is $E/\langle a\rangle$. The case $c = (0, b)$ is obtained by interchanging the factors and gives two copies of $F/\langle b\rangle$.

For $c = (a, d)$ with $d \notin \{0, b\}$, the second set consists of two copies of E, indexed by the roots of $2y = b + d$ modulo sign. The operation $y \mapsto -y + b$ exchanges the two pairs. Indeed it could preserve a pair only if $-y + b = y$ or $-y + b = -y$, requiring respectively $2y = b$ or $b = 0$, both impossible. Thus σ exchanges the two elliptic curves, and their image is one curve isomorphic to E. The fourth family is symmetric and gives one copy of F.

There remain eight nonzero c for which both components of $c_0 + c$ are nonzero. The map σt_c is then free, so it adds no curve. The list has $1+1+1+2+2+8 = 15$ elements and is exhaustive. □

6.2. **The numerical representation.**

Corollary 6.2. *The kernel of the translation action on $H^2(S, \mathbb{Q})$ is $\{0, c_0\}$. With $\overline{T} = T/\langle c_0\rangle$, the faithful representation is*

$$H^2(S, \mathbb{Q}) \simeq \mathbb{Q}^3 \oplus \bigoplus_{\substack{\chi \in \widehat{\overline{T}} \\ \chi \neq 1}} \mathbb{Q}_\chi.$$

Every nonidentity element of $\overline{T}$ has trace two and six-dimensional invariant subspace. The map $\bar{t}_{c_0}$ is numerically trivial, although it is not the identity automorphism.

Proof. The pullback identifies $H^2(S, \mathbb{Q})$ with the σ-invariants in (4). In the $H^2(A)$ summand only the two factor classes survive. Among the sixteen exceptional characters, precisely the eight with $\chi(c_0) = 1$ survive. They form the regular representation of $\overline{T}$. This proves the decomposition and its kernel. Alternatively, the special fixed locus has Euler characteristic $4 \cdot 2 + 4 = 12$, giving trace ten; all other fixed loci have Euler characteristic four, giving trace two. Since $h^{2,0}(S) = 0$, the rational second cohomology is algebraic by the integral $(1, 1)$ theorem, so triviality on rational cohomology is triviality on the numerical divisor space. Its integral lattice is torsion-free; a rationally trivial integral action is the identity. □

7. General K3 involutions and the Kummer distinction

The explicit construction illustrates a general local dichotomy but does not remove the hypotheses of that dichotomy.

Proposition 7.1. *A nonidentity symplectic involution of a complex K3 surface has eight isolated fixed points. An antisymplectic involution has only smooth fixed curves, or has empty fixed locus.*

Proof. Finite-order analytic actions are locally linearizable by averaging coordinates. At a fixed point the eigenvalues are ± 1. A symplectic action has determinant one. If both eigenvalues are $+1$, linearization makes the automorphism locally, hence globally, the identity. Thus a nonidentity symplectic involution has derivative $-I$ at every fixed point. Its holomorphic Lefschetz number on $\mathcal{O}$ is $1 + 1 = 2$; each such point contributes $1/\det(I - (-I)) = 1/4$. There are eight points.

An antisymplectic action has determinant -1, so the eigenvalues are $1, -1$. Its fixed set is locally the smooth $+1$ eigendirection. Different fixed components cannot cross, because the local fixed set is a single smooth curve. □

The holomorphic Lefschetz theorem is used here in its standard surface form, not reproved. The product calculation of the eight translation fixed points in Proposition 3.1 did not need it.

Proposition 7.2. *The resolution of the quotient of a K3 surface by a symplectic involution is K3. It need not be Kummer.*

Proof. The quotient has eight A_1 singularities. The local resolution extends the invariant two-form without zeros, exactly as in Section 2. Invariant $H^1(\mathcal{O})$ vanishes, proving the K3 assertion.

For the second assertion, projective K3 surfaces with a symplectic involution and Picard number nine exist; the families and their lattices are described by van Geemen and Sarti [3]. The quotient resolution has the same transcendental rank: the degree-two quotient correspondence identifies the transcendental rational Hodge structures, with the intersection form rescaled. Indeed a symplectic involution acts trivially on the transcendental part, since its negative eigenspace is of type $(1, 1)$ and therefore algebraic. Consequently both K3 surfaces have Picard number nine. Every projective Kummer surface has Picard number at least seventeen: the sixteen independent exceptional classes are algebraic, and a pulled-back polarization adds one more independent positive-square class. The Picard-nine quotient is therefore not Kummer. □

The use of the cited Picard-nine family is essential to this counterexample. The explicit product construction, by contrast, has enough additional geometry to identify its translation quotients as Kummer surfaces in Proposition 3.2.

8. The Fermat quartic example corrected

Consider the smooth quartic

$$Y = \{x_0^4 + x_1^4 + x_2^4 + x_3^4 = 0\} \subset \mathbb{P}^3$$

and the projective involution

$$\iota[x_0 : x_1 : x_2 : x_3] = [-x_0 : -x_1 : x_2 : x_3].$$

A projective fixed point is an eigenline of the diagonal matrix. The two eigenspaces are the lines $L_- :\ x_2 = x_3 = 0$ and $L_+ :\ x_0 = x_1 = 0$. Their intersections with Y are respectively

$$[z : 1 : 0 : 0], \qquad [0 : 0 : z : 1], \qquad z^4 = -1.$$

The polynomial $z^4 + 1$ has four distinct roots because it has no common root with $4z^3$. Thus ι has exactly eight fixed points.

The holomorphic two-form of a smooth quartic is the residue of the standard homogeneous three-form divided by the defining polynomial. The latter is invariant and the diagonal transformation has determinant $+1$, so ι is symplectic. Its quotient resolution is K3 and is not a free Enriques quotient. This corrects the earlier example without changing the free Lieberman construction.

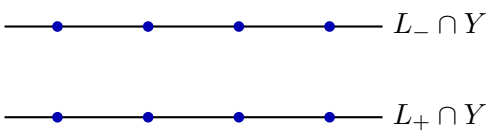


FIGURE 4. The diagram records two disjoint projective eigenlines and their eight intersection points. It is an incidence schematic rather than a spatial embedding of the quartic surface.

9. Projective and lattice consequences

For a complex K3 surface, projectivity is equivalent to the existence of an integral $(1,1)$ class with positive square [2]. Merely requiring a nonzero Néron–Severi group is insufficient: a nonprojective K3 surface can have a negative-definite or degenerate Néron–Severi lattice. If L is ample, a sufficiently high power is very ample; L itself need not define an embedding. In particular the existence of a polarization must not be turned into an arbitrary quartic model.

The Kummer quotient $A/\langle -1\rangle$ can be singular even when an associated linear system realizes it projectively. Its smooth resolution contains sixteen (-2)-curves. A divisor giving a contraction has zero intersection with those curves, and is nef rather than ample on the resolution. The behavior of a product principal polarization also differs from that of an indecomposable principally polarized abelian surface. None of the arguments in this paper requires a quartic embedding of the product Kummer surface.

For completeness, the standard integral lattice identifications used for comparison are recorded in Appendix B. Our rational character decomposition does not require their integral classification. This separation prevents an unnoticed finite-index issue from entering the fixed-locus calculation.

10. Exact checks and reproducibility

The file `checks/verify.py` uses only Python's standard library. It models T as $\mathbb{F}_2^4$ and G as $\mathbb{F}_2^5$, lists all 31 nonidentity elements, and checks the counts in Theorem 1.1. It computes traces independently from the permutation of the sixteen node labels and from the character table. For each of the nine possible c_0, it checks the 15-case descent table and the multiplicity formula on S.

The geometric proof applies to every elliptic curve, whereas the program checks its finite combinatorial shadow. It does not solve equations numerically on a chosen elliptic curve, infer freeness from sampling, or claim a formalization in a proof assistant. The Fermat root check is exact: the Euclidean algorithm for $z^4 + 1$ and $4z^3$ gives constant greatest common divisor.

Appendix A. Local quotient coordinates and fixed curves

It is useful to keep three maps separate. On A the inversion has isolated fixed points. On the blown-up surface $\widetilde{A}$ its lift fixes each exceptional curve pointwise. On X that exceptional curve is the branch image and has self-intersection -2. These different fixed sets belong to different surfaces.

In the first blowup chart $z = u, w = uv$, inversion is $(u, v) \mapsto (-u, v)$ and the quotient is $(t, v) = (u^2, v)$. In the second chart $z = sw, w = w$, the quotient is $(s, q) = (s, w^2)$. The transition is

$$s = v^{-1}, \qquad q = tv^2.$$

The exceptional curve has normal transition $q = tv^2$, the transition of $\mathcal{O}_{\mathbb{P}^1}(-2)$ under the corresponding coordinate convention. The two-form patches agree because

$$\frac{1}{2}\, dt \wedge dv = -\frac{1}{2}\, dq \wedge ds.$$

This is the complete local resolution of the ordinary double point.

For the reflection $(z, w) \mapsto (z, -w)$, the quotient coordinates give $(t, v) \mapsto (t, -v)$ in the first chart and $(s, q) \mapsto (-s, q)$ in the second. The two exceptional fixed points are $v = 0$ and $s = 0$. They lie on the strict transforms of the two coordinate axes, respectively. This proves both disjointness of the eight rational fixed curves and the absence of additional isolated points for r.

Appendix B. Integral lattices and the rational calculation

The following are standard integral lattice theorems; they are background, not claimed as new proofs here [2, 3, 4]:

$$\begin{aligned}
H^2(X, \mathbb{Z}) &\simeq U^{\oplus 3} \oplus E_8(-1)^{\oplus 2}, \\
H^2(X, \mathbb{Z})^{\nu} &\simeq U^{\oplus 3} \oplus E_8(-2) && \text{for a symplectic involution } \nu, \\
H^2(X, \mathbb{Z})^{-\nu} &\simeq E_8(-2), \\
\mathrm{Num}(S) &\simeq U \oplus E_8(-1), \\
H^2(X, \mathbb{Z})^{\sigma} &\simeq U(2) \oplus E_8(-2) && \text{for a free involution } \sigma.
\end{aligned}$$

The ranks in these formulas are 22, 14, 8, 10 and 10. They agree with the independently derived traces in Corollary 5.2. The factor two for the free quotient also agrees with (3).

For a projective K3 surface, let $T_X = \mathrm{NS}(X)^{\perp}$. Orthogonality gives

$$H^2(X, \mathbb{Q}) = \mathrm{NS}(X)_{\mathbb{Q}} \oplus T_{X,\mathbb{Q}}.$$

Over $\mathbb{Z}$, the inclusion $\mathrm{NS}(X) \oplus T_X \subset H^2(X, \mathbb{Z})$ usually has a nontrivial finite cokernel. Discriminant forms supply the gluing. Thus a rational direct-sum calculation cannot be promoted to an integral orthogonal decomposition without checking that gluing.

Similarly, (4) is a rational G-module statement. It makes no assertion that the sixteen exceptional classes and a chosen integral copy of $H^2(A, \mathbb{Z})$ form the entire integral K3 lattice. Their primitive closure contains additional classes.

Appendix C. An explicit enumeration of the descent cases

Choose bases for $E[2]$ and $F[2]$ and write their elements as $0, 1, 2, 3$ with addition given by bitwise exclusive-or. Because any nonzero vector in $\mathbb{F}_2^2$ can be sent to any other by a basis change, it suffices for the combinatorial list to take $a = b = 1$. The special cases are

$$(1,1), \quad (1,0), \quad (0,1), \quad (1,2),\ (1,3), \quad (2,1),\ (3,1).$$

They give respectively four rational curves, two elliptic curves, two elliptic curves, and one elliptic curve for each of the last four elements. The remaining nonzero elements are

$$(0,2), (0,3), (2,0), (3,0), (2,2), (2,3), (3,2), (3,3).$$

For each of them both components of $c + c_0$ are nonzero. This list also shows directly why there are eight curve-free cases.

For a character $\chi : T \to \{\pm 1\}$, the projector

$$P_{\chi} = \frac{1}{16} \sum_{c \in T} \chi(c) t_c^*$$

has rank three for the trivial character and rank one for each nontrivial character on the reflection-positive part of $H^2(X)$. On the reflection-negative part the trivial character has rank four and all other projectors have rank zero. Imposing σ-invariance retains exactly those positive characters with $\chi(c_0) = 1$. This gives an explicit projection formula for every summand used in Corollary 6.2.

Appendix D. Revision record and scholarly boundaries

The earlier review's broad surface background has been retained in corrected form where it is needed for this construction. Unsupported general classifications and the false free Fermat example have been removed or corrected. The paper supplies complete fixed-locus arguments for the specified group, the quotient identification, the descent table, and the rational representations. Integral lattice classification, holomorphic Lefschetz theory, and general projectivity criteria are explicitly cited standard inputs.

The article does not classify all automorphisms of K3 or Enriques surfaces, all Enriques quotients of a product Kummer surface, or all Kummer models. Classical Lieberman constructions are credited as such. The revised proofs and exact checks do not by themselves establish that any result here is a new major conjecture resolution.

Declarations and availability

Competing interests. The authors declare no competing interests. **Funding.** No specific funding is reported in the supplied source. **Ethics.** This work involves no human participants or animals. **Authorship.** The published author order and corresponding author are preserved. Pallab Nandi's IISER association is historical, as specified in the earlier version. **Computational and editorial assistance.** AI assistance was used for source review, derivation checks, code and preparation of this revision. It is not an author. The accompanying provenance file records the scope of those checks; the article is not Lean-formalized. **Acknowledgements.** The exposition builds on the work of Lieberman, Nikulin and the authors cited below. No endorsement by them is implied.

Repository and release. `https://github.com/creelie/k3-involutions`. Release v2.0.0.

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Electro–Gravitational Space Propulsion Laboratory (EGSPL), India
Email address: `itsdeep@live.com`

Former student, Indian Institute of Science Education and Research Kolkata, India
Email address: `pallabnandi995@gmail.com`